Multivalent Ions

- Some elements can form more than one ion
- There are 2 systems in use to name these ions

IUPAC – International Union of Pure & Applied Chemists

Ion	Stock Name (old method)	IUPAC Name (new method)
Fe^{2+}	Ferrous	Iron (II)
Fe^{3+}	Ferric	Iron (III)
Cu^{1+}	Cuprous	Copper (I)
Cu^{2+}	Cupric	Copper (II)
Zn^{2+}	--------------	Zinc
Ag^{+}	--------------	Silver
Co^{2+}	Cobaltous	Cobalt (II)
Co^{3+}	Cobaltic	Cobalt (III)
As^{3+}	Arsenious	Arsenic (III)
As^{5+}	Arsenic	Arsenic (V)
Cd^{2+}	--------------	Cadmium
Sn^{2+}	Stannous	Tin (II)
Sn^{4+}	Stannic	Tin (IV)
Sb^{3+}	Antimonous	Antimony (III)
Sb^{5+}	Antimonic	Antimony (V)
Au^{1+}	Aurous	Gold (I)
Au^{3+}	Auric	Gold (III)
Hg^{1+}	Mercurous	Mercury (I)
Hg^{2+}	Mercuric	Mercury (II)
Pb^{2+}	Plumbous	Lead (II)
Pb^{4+}	Plumbic	Lead (IV)

Note: The ion with the lower valence has an “ous” suffix and the higher one has an “ic” suffix.

Multivalent Ioinc Bonding

- Same rules apply as in ionic bonding
- Only this time the old **stock name** may be used and if the **IUPAC** system is used, the charge on the metallic ion must be shown in roman numerals.
- The numbers in roman numerals refers to the charge on the metallic ions

Example:

1. SnF_2 = stannous fluoride *or* tin (II) fluoride
2. Gold (III) Oxide = $Au^{3+}\ O^{2-}$ = Au_2O_3
3. Lead (IV) Oxide = $Pb^{4+}\ O^{2-}$ = Pb_2O_4 = PbO_2 (*reduce*)

<u>Example 1)</u>

Write the chemical name for BeH_2

<u>Solution 1)</u>

1. Write the names of the cation and the anion. The anion must end in the suffix "-ide"

   beryllium hydride

2. Since beryllium has a fixed valence, we are finished.

<u>Example 2)</u>

Write the chemical name for FeS

<u>Solution 2)</u>

1. Write the names of the cation and the anion. The anion must end in the suffix "-ide"

   iron sulfide

2. Since iron has more than one valence, use the subscripts to determine the charges on the ions.

   Fe^{3+} S^{3-} *but the charge on S is 2-!*
   Fe^{2+} S^{2-} *iron (II) sulfide*

Give the correct formula:

Chemical Name	Ions Formed		Chemical Formula
Cobalt (II) chloride	Co^{2+}	Cl^{1-}	$CoCl_2$
Tin (IV) nitride	Sn^{4+}	N^{3-}	Sn_3N_4
Zinc iodide	Zn^{2+}	I^{1-}	ZnI_2
Plumbous chloride	Pb^{2+}	Cl^{1-}	$PbCl_2$
Silver Bromide	Ag^{+}	Br^{1-}	$AgBr$
Antimony (III) phosphide	Sb^{3+}	P^{3-}	SbP
Mercuric oxide	Hg^{2+}	O^{2-}	HgO
Cupric bromide	Cu^{2+}	Br^{1-}	$CuBr_2$
Ferric sulphide	Fe^{3+}	S^{2-}	Fe_2S_3
Mercurous iodide	Hg^{+}	I^{1-}	HgI

Name the following compounds:

Chemical Name	IUPAC Name	Stock Name
CuO	Copper (II) Oxide	Cupric Oxide
SnF_4	Tin (IV) fluoride	Stannic fluoride
$PbCl_4$	Lead (IV) chloride	Plumbic chloride
$AsCl_5$	Arsenic (V) chloride	Arsenic chloride
$AgBr$	Silver bromide	-----
CdI_2	Cadmium Iodide	-----
ZnO	Zinc Oxide	-----
Sb_2O_3	Antimony (III) Oxide	Antimonious Oxide
Fe_2O_3	Iron (III) Oxide	Ferric Oxide
$AuBr_3$	Gold (III) bromide	Auric bromide

Write the chemical formula for the following binary ionic compounds:

a) Calcium chloride $CaCl_2$
b) Potassium oxide K_2O
c) Lithium nitride Li_3N
d) Cadmium oxide CdO
e) Silver chloride $AgCl$
f) Mercury (I) sulfide Hg_2S
g) Barium fluoride BaF_2
h) Sodium bromide $NaBr$
i) Zinc iodide ZnI_2
j) Copper (II) chloride $CuCl_2$
k) Gold (I) iodide AuI
l) Nickel (II) sulfide NiS
m) Aluminum hydride AlH_3
n) Iron (III) chloride $FeCl_3$
o) Potassium nitride K_3N
p) Magnesium nitride Mg_3N_2
q) Tin (IV) fluoride SnF_4
r) Iron (III) sulfide Fe_2S_3
s) Cobalt (II) bromide $CoBr_2$
t) Manganese (II) sulfide MnS
u) Lead (IV) oxide PbO_2
v) Cadmium chloride $CdCl_2$
w) Strontium oxide SrO
x) Chromium (II) fluoride CrF_2
y) Lithium sulfide Li_2S
z) Silver oxide Ag_2O

Give the chemical name for the following binary ionic compounds. Include both the IUPAC and stock names, where applicable.

a) FeS Iron (II) / Ferrous sulfide
b) $CrCl_3$ Chromium (III) / Chromic chloride
c) MgO Magnesium oxide
d) $AlCl_3$ Aluminum chloride
e) LiF lithium fluoride
f) K_2S potassium sulfide
g) $ZnBr_2$ zinc bromide
h) FeO Iron (II) / Ferrous oxide
i) Fe_2O_3 Iron (III) / Ferric oxide
j) NiF_2 nickel (II) / nickelous fluoride
k) CuS copper(II) / cupric sulfide
l) Hg_2O mercury (I) / mercurous oxide
m) AlN aluminum nitride
n) $BaBr_2$ barium bromide
o) Ag_2S silver sulfide
p) $BiCl_3$ bismuth (III) / bismous chloride
q) SnO Tin (II) / stannous oxide
r) CdS cadmium sulfide
s) MnO_2 manganese (IV) / manganic oxide
t) SrH_2 strontium hydride
u) AuF gold (I) / aurous fluoride
v) UF_4 uranium (IV) / uranous fluoride
w) Co_2O_3 cobalt (III) / cobaltic oxide
x) K_3N potassium nitride
y) PbI_2 lead (II) / plumbous iodide
z) SnO_2 tin (IV) / stannic oxide

Polyatomic Anions (Negative Ions)

Polyatomic ions consist of **groups of atoms** that carry an overall negative charge. Since these are very common and have their own special name, you are required to learn them. Treat polyatomic ions as a "package deal".

-1	-2	-3
$C_2H_3O_2^-$ Acetate CH_3COO^-	CO_3^{2-} Carbonate	PO_4^{3-} Phosphate
ClO_3^- Chlorate	SO_4^{2-} Sulphate	
BrO_3^- Bromate	CrO_4^{2-} Chromate	
IO_3^- Iodate	$C_2O_4^{2-}$ Oxalate	
HCO_3^- Bicarbonate Hydrogen Carbonate	$Cr_2O_7^{2-}$ Dichromate	
NO_3^- Nitrate		
MnO_4^- Permanganate		
SCN^- Thiocyanate		
CN^- Cyanide		
OH^- Hydroxide		
HSO_4^- Hydrogen Sulphate		
HS^- Hydrogen Sulfide		

Complex Oxyanions: contain oxygen

There is only one **polyatomic cation** (positive polyatomic ion)

Ammonium NH_4^+

Not to be confused with **Ammonia NH_3**

Sample Problems:

Name	Ions	Chemical Formula
Sodium Carbonate	Na^{1+} CO_3^{2-}	Na_2CO_3
Copper (II) Sulphate	Cu^{2+} SO_4^{2-}	$CuSO_4$
Copper (II) Bromate	Cu^{2+} BrO_3^-	$Cu(BrO_3)_2$

****ENCLOSE THE ENTIRE POLYATOMIC ION IN BRACKETS WHEN THE SUBSCRIPT IS GREATER THAN 1**

Polyatomic Ions

Name	Ions	Chemical Formula
Silver acetate	Ag^+ $C_2H_3O_2^-$	$AgC_2H_3O_2$
Potassium permanganate	K^+ MnO_4^-	$KMnO_4$
Tin (II) dichromate	Sn^{+2} $Cr_2O_7^{-2}$	$SnCr_2O_7$
Zinc hydroxide	Zn^{+2} OH^-	$Zn(OH)_2$
Magnesium oxalate	Mg^{+2} $C_2O_4^{-2}$	MgC_2O_4
Barium oxalate	Ba^{+2} $C_2O_4^{-2}$	BaC_2O_4
Aluminum acetate	Al^{+3} $C_2H_3O_2^-$	$Al(C_2H_3O_2)_3$
Copper (II) hydroxide	Cu^{+2} OH^-	$Cu(OH)_2$
Cupric nitrate	Cu^{+2} NO_3^-	$Cu(NO_3)_2$
Lead (II) sulphate	Pb^{2+} SO_4^{-2}	$PbSO_4$
Lead (II) nitrate	Pb^{2+} NO_3^-	$Pb(NO_3)_2$
Mercury (I) phosphate	Hg^+ PO_4^{-3}	Hg_3PO_4
Potassium thiocyanate	K^+ SCN^-	$KSCN$
Sodium Oxalate	Na^+ $C_2O_4^{-2}$	$Na_2C_2O_4$
Gold (III) cyanide	Au^{+3} CN^-	$Au(CN)_3$

Sodium permanganate	Na^{+} MnO_4^{-}	$NaMnO_4$
Mercury (II) nitrate	Hg^{2+} NO_3^{-}	$Hg\ (NO_3)_2$
Chromium (III) oxalate	Cr^{3+} $C_2O_4^{-2}$	$Cr_2(C_2O_4)_3$
Cobalt (II) acetate	Co^{2+} $C_2H_3O_2^{-}$	$Co(C_2H_3O_2)_2$
Ammonium dichromate	NH_4^{+} $Cr_2O_7^{2-}$	$(NH_4)_2\ Cr_2O_7$
Barium carbonate	Ba^{2+} CO_3^{2-}	$BaCO_3$
Lead (IV) sulphate	Pb^{4+} SO_4^{-2}	$Pb(SO_4)_2$
Potassium bicarbonate	K^{+} HCO_3^{-}	$KHCO_3$
Copper (II) sulphate	Cu^{2+} SO_4^{2-}	$CuSO_4$
Calcium carbonate	Ca^{2+} CO_3^{2-}	$CaCO_3$
Iron (II) sulfate	Fe^{2+} SO_4^{2-}	$FeSO_4$
Copper (II) nitrate	Cu^{2+} NO_3^{-}	$Cu(NO_3)_2$
Sodium bicarbonate	Na^{+} HCO_3^{-}	$NaHCO_3$
Ferrous phosphate	Fe^{2+} PO_4^{3-}	$Fe_3\ (PO_4)_2$
Sodium thiocyanate	Na^{+} SCN^{-}	$NaSCN$
Lithium Sulfate	Li^{+} SO_4^{2-}	Li_2SO_4
Ammonium phosphate	NH_4^{+} PO_4^{3-}	$(NH_4)_3PO_4$
Calcium Sulfate	Ca^{2+} SO_4^{2-}	$CaSO_4$
Sodium hydroxide	Na^{+} OH^{-}	$NaOH$
Stannous dichromate	Sn^{2+} $Cr_2O_7^{2-}$	$SnCr_2O_7$
Plumbous acetate	Pb^{2+} $C_2H_3O_2^{-}$	$Pb(C_2H_3O_2)_2$
Mercury (II) phosphate	Hg^{2+} PO_4^{3-}	$Hg_3(PO_4)_2$
Ammonium oxalate	NH_4^{+} $C_2O_4^{2-}$	$(NH_4)_2C_2O_4$
Iron (III) hydroxide	Fe^{3+} OH^{-}	$Fe(OH)_3$
Magnesium carbonate	Mg^{2+} CO_3^{2-}	$MgCO_3$

Nomenclature for Oxyanions

O's	Formula	Name	Style
One more	ClO_4^-	**Per**chlorate	Per………..ate
------------	ClO_3^-	Chlor**ate**	……………….ate
One fewer	ClO_2^-	Chlor**ite**	……………….ite
Two fewer	ClO^-	**Hypo**chlor**ite**	Hypo………ite

Example:

IO_2^- **Iodate** (parent name) becomes **Iodite** (*one fewer oxygen*)

KNO_2 Potassium Nitrite

Chemical Equation	Name
SO_3^{2-}	Sulfite
BrO_3^-	Bromate
BrO_2^-	Bromite
BrO^-	Hypobromite
IO_4^-	Periodate
IO_3^-	Iodate
IO_2^{--}	Iodite

Practice Problems

Chemical Equation	Name
Na_2CO_3	Sodium carbonate
$Pb(NO_3)_4$	Lead (IV) nitrate
$Sn(ClO)_2$	Tin (II) hypochlorite
Ag_2SO_3	Silver sulfite
$Mg_3(PO_2)_2$	Magnesium hypophosphite
$Ca(NO_2)_2$	Calcium nitrite
$AlPO_4$	Aluminum phosphate
KIO	Potassium hypoiodite
$Zn(BrO_4)_2$	Zinc perbromate
CuClO	Copper (I) hypochlorite
$Hg(IO_4)_2$	Mercuric periodate
$Ca(BrO)_2$	Calcium hypobromite
K_2SO_3	Potassium sulfite
$Ni(ClO_2)_2$	Nickel (II) chlorite
$Al_2(CO_3)_3$	Aluminum carbonate
$Sn(IO_4)_2$	Stannous periodate
$Au(BrO_3)_3$	Gold (III) bromate
$CuSO_4$	Copper (II) sulphate
$NaNO_2$	Sodium nitrite
$Pb(IO_4)_4$	Lead (IV) periodate

Oxy-acids

Oxy – meaning they contain oxygen **Acid** – meaning they contain hydrogen

Naming:
All oxy-acids get their name logically from their parent oxyanions by changing the ending from **"-ate"** to **"-ic acid"**

-ic Acids

H	**H_2**	**H_3**
$HClO_3$ Chlor**ic** Acid	H_2SO_4 Sulphur**ic** Acid	H_3PO_4 Phosphor**ic** Acid
$HBrO_3$ Brom**ic** Acid	H_2CO_3 Carbon**ic** Acid	
HNO_3 Nitr**ic** Acid		
HIO_3 Iod**ic** Acid		

-ous Acids *Oxyacids with one less oxygen than their parents*

H	**H_2**	**H_3**
$HClO_2$ Chlor**ous** Acid	H_2SO_3 Sulphur**ous** Acid	H_3PO_3 Phosphor**ous** Acid
$HBrO_2$ Brom**ous** Acid	------------	
HNO_2 Nitr**ous** Acid		
HIO_2 Iod**ous** Acid		

Hypo- -ous Acids *Oxyacids with two less oxygen than their parents*

H	**H_2**	**H_3**
HClO **Hypo**chlor**ous** Acid	------------	H_3PO_2 **Hypo**phosphor**ous** Acid
HBrO **Hypo**brom**ous** Acid	------------	

HIO **Hypo**iod**ous** Acid		

Per---ic Acids *Oxyacids with one more oxygen than their parents*

H	**H_2**	**H_3**
$HClO_4$ **Per**chlor**ic** Acid	------------	------------
$HBrO_4$ **Per**brom**ic** Acid	------------	

HIO_4 **Per**iod**ic** Acid		

Binary Acids

The following is a list of common binary acids that must me memorized:

Name	Formula
Hydrochloric acid *or* hydrogen chloride	HCl
Hydrofluoric acid *or* hydrogen fluoride	HF
Hydroiodic acid *or* hydrogen iodide	HI
Hydrobromic acid *or* hydrogen bromide	HBr
Hydrosulfuric acid *or* hydrogen sulfide	H_2S
Hydroselenic acid *or* hydrogen selenide	H_2Se

Note – these do not contain oxygen

Practice Problems

Chemical Equation	Name
HCl	Hydrochloric acid
H_2SO_4	Sulfuric acid
HNO_2	Nitrous acid
HBr	Hydrobromic acid
$HClO_4$	Perchloric acid
H_2Se	Hydroselenic acid
$HBrO_3$	Bromic acid
H_2CO_3	Carbonic acid
H_3PO_4	Phosphoric acid
HIO	Hypoiodous acid
$HC_2H_3O_2$	Acetic acid
HF	Hydrofluoric acid
H_3PO_3	Phosphorous acid
$HClO_2$	Chlorous acid
HNO_3	Nitric acid
HBr	Hydrobromic acid
HI	Hydroiodic acid
H_2S	Hydrosulphuric acid
H_2SO_3	Sulphurous acid
HIO_2	Iodous acid
HIO_4	Periodic acid
HNO_2	Nitrous acid
H_2CO_3	Carbonic acid